5. $(x-1)(y+1)=C$ 6.(i) $\log y=\frac{-\cos 2x}{2}+C$ (ii) $\frac{e^{ax}}{a}=\frac{-e^{by}}{b}+C$

7. $(y-b)^2=(x-a)^2+b^2-a^2$

Exercise 4.3

1. $x=Ce^{\frac{y}{x}}$ 2. $x+y=Ke^{\frac{-2x}{x+y}}$ 3. $y+\sqrt{x^2+y^2}=x^2C$

4. $3y^2-4yx+3x^2=x^3C$ 5. $\left(xy-y^2\right)x=C$ 6. $y\sqrt{y^2-x^2}=2\sqrt{3}\,x^5$

7. $y=ce^{\frac{x^2}{2y^2}}$

Exercise 4.4

1. $\frac{y}{x}=x+C$ 2. $ye^{\sin x}=e^{\sin x}(\sin x-1)+C$ 3. $x^2y=\frac{x^6}{6}+C$

4. $y\left(1+x^3\right)=x+\frac{x^3}{3}+C$ 5. $xy=e^x\left(x^2-2x+2\right)+C$ 6. $y\sec x=\frac{1}{2}\left(x+\frac{\sin 2x}{2}\right)+C$

7. $y\sec^2 x=\sec x-2$ 8. $x^2e^x-2xe^x+2e^x+c$ 9. ₹ 2,22,550

Exercise 4.5

1. $y=Ae^{2x}+Be^{4x}$ 2. $y=(Ax+B)e^{2x}$ 3. $y=e^{-x}(A\cos\sqrt{2}\,x+B\sin\sqrt{2}\,x)$

4. $y=(Ax+B)e^{kx}$ 5. $y=\frac{e^{-3x}}{12}+\frac{e^{5x}}{20}$ 6. $y=Ae^{\frac{1}{2}x}+Be^{\frac{-3}{2}x}+\frac{e^{2x}}{21}$

7. $y=A\cos 4x+B\sin 4x$ 8. $y=e^x-\frac{3}{2}e^{2x}+\frac{e^{3x}}{2}$

9. $y=Ae^{-3x}+Be^{2x}+\frac{e^{3x}}{6}-\frac{x}{5}e^{-3x}$ 10. $y=(Ax+B)e^{5x}+2x^2e^{5x}+\frac{1}{5}$

11. $y=Ae^{\frac{-3}{2}x}+Be^{\frac{-5}{2}x}+xe^{-\frac{3}{2}x}$ 12. $y=Ae^{2x}+Be^{\frac{-7}{3}x}+xe^{2x}$

13. $P=Ae^{-4t}+Be^{2t}+2$

Exercise 4.6

1	2	3	4	5	6	7	8	9	10	11	12	13
(a)	(d)	(a)	(b)	(a)	(a)	(d)	(c)	(a)	(c)	(b)	(a)	(b)
14	15	16	17	18	19	20	21	22	23	24	25	
(c)	(d)	(a)	(d)	(a)	(b)	(d)	(a)	(a)	(d)	(c)	(a)	

Miscellaneous Problems

1. $p=Ae^{-4t}+Be^{2t}+3$ 2. $x^2\frac{d^2y}{dx^2}-2x\frac{dy}{dx}+2y=0$ 3. $e^x\left(x^2-2x+2\right)+\log y=c$

4. $x\left(1+\frac{3y^2}{x^2}\right)^{\frac{1}{3}}=c$ 5. $yx^2=\frac{x^6}{6}+c$ 6. $cm^2=2(m+6)$

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